Name = Sarfraaz Shaikh
Enrollment Number = A71004824 54
Department = AIIT
Course = BCA

Q1. Explain in detail about the AWS Environment
What is AWS?
AWS (Amazon Web Services) is a cloud computing platform provided by Amazon that offers a wide range of services such as computing power, storage, networking, databases, machine learning, and security, all available through the internet (the "cloud").

Main components of the AWS Environment
Compute Services
These provide the processing power to run applications.

Example: Amazon EC2 (Elastic compute cloud), AWS Lambda, Elastic Beanstalk

Storage Services
AWS provides various storage options for

different needs.

Example: Amazon S3 (Simple Storage Service) for object storage

Amazon EBS (Elastic Block Store) for block storage

Amazon Glacier for archival storage

Networking Services
AWS manages network connections and content delivery.

Example: Amazon VPC (Virtual Private Cloud) for creating private networks

Route 53 for DNS services

CloudFront for content delivery

Database Services

Example: Amazon RDS (Relational Database

Service)

DynamoDB for NoSQL databases

Redshift for data warehousing

Security & Identity Management

Example: IAM (Identity and Access Management) to control access

AWS Shield and WAF for security protection

Monitoring and Management

Example: CloudWatch (monitoring resources)

CloudTrail (logging API calls)

Advantages of AWS Environment
Scalability: Easily increase or decrease resources as needed

Cost-Effective: Pay only for what you use

Reliability: High uptime and data durability

Global Reach: Data centers in multiple regions around the world

Security: End-to-end encryption and compliance certifications

Q2. What is EC2? How does it help the AWS mechanism?

Meaning of EC2

EC2 (Elastic compute cloud) is a virtual server in the AWS cloud.

It allows users to run applications on Amazon's cloud infrastructure using virtual machines (called Instances).

Key Features of EC2

Elasticity: Easily increase or decrease the number of instances according to demand.

Customizable: Choose CPU, RAM, storage, and operating system as per your needs.

Secure: Integrated with IAM roles, security groups, and key pairs.

Cost Options: Pay as you go, spot instances, or reserved instances.

Automation: Can be scaled automatically using Auto Scaling.

How EC2 Helps the AWS Mechanism
Provides the core computing power for AWS services.

Hosts applications, websites, databases, and microservices.

Integrates with EBS (for storage), VPC (for networking), and CloudWatch (for monitoring).

Enables AWS to offer Infrastructure as a Service (IaaS) to users.

In short, EC2 acts as the "engine" that runs

most AWS workloads.

Q3. Explain the role of EBS (Elastic Block Storage) in detail

What is EBS?

Amazon EBS (Elastic Block Store) is a block-level storage service designed for use with EC2 instances.

It provides persistent storage - meaning your data remains available even after you stop or restart an EC2 instance.

Features of EBS

Persistent Storage: Data is not lost when the instance stops or restarts.

High Performance: Designed for both transactional and throughput-intensive workloads.

Scalable: You can increase storage capacity anytime without downtime.

Backup: Supports EBS Snapshots (stored in

Amazon S3).

Encrypted: Data is encrypted at rest and in transit.

Availability: Data is automatically replicated within its Availability Zone.

Types of EBS Volumes

Type	Description	Use Case
General Purpose SSD (gp3/gp2)	Balanced performance and cost	Everyday workloads
Provisioned IOPS SSD (io2/io1)	High performance for intensive applications	Databases, OLTP
Throughput Optimized HDD (st1)	High throughput	Big data, log processing
Cold HDD (sc1)	Low-cost storage	Infrequent access

Role of EBS in AWS

Acts as a hard drive for EC2 instances.

Stores operating systems, databases, and application data.

Provides quick recovery via EBS snapshots.

Ensures data reliability through automatic replication.

Example Scenario
When you launch an EC2 instance, you attach an EBS volume to it.
If the EC2 instance is like a "computer," then the EBS volume acts like its "hard disk."

Summary Table:

Component	Full Form	Function
EC2	Elastic Compute Cloud	Provides virtual servers for computing
EBS	Elastic Block Store	Provides persistent storage for EC2 instances
AWS Environment	Amazon Web Services Environment	Full cloud ecosystem of compute, storage, networking, and more

Would you like me to make a diagram that visually explains how EC2 and EBS fit into the AWS environment?

Is this conversation helpful so far?